Debjyoti Chatterjee

Ph.D. candidate, University of Texas at Austin

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Education

Ph.D. candidate in Electrical and Computer Engineering **Expected September 2026**
The University of Texas at Austin, TX, GPA: 3.87/4.0
Advisor: Dr. Brian Johnson
Thesis: Current-constrained Control and Stability Assessment of Grid-forming (GFM) Inverter Systems

M.S. in Electrical and Computer Engineering **Dec 2022**
The University of Texas at Austin, TX, GPA: 3.92/4.0
Thesis: From Grid-following to Grid-forming: Modeling, Control & Applications to Inverter-based Resources

B. Tech. in Electrical and Electronics Engineering **May 2020**
National Institute of Technology, Tiruchirappalli, India, GPA: 9.26/10.0

Areas of Expertise

- Power electronic interfaces for PV/BESS, data centers
- Dynamic system modeling and data-driven analysis
- Stability analysis of large scale systems
- Control system design and validation
- Control and optimization of energy systems
- Feedback control of complex physical systems

Relevant Coursework

Control of Power Electronics, Power System Operation and Control, Power System Analysis with Renewable Energy Sources, Fundamentals of Power Electronics, Machines and Drives, Intro to Modern Control, Power Quality and Harmonics

Technical Skills

- **Software Platforms**: MATLAB/Simulink, PSCAD, PSSE, PLECS, EMTP, Opal-RT
- **Microcontrollers**: TI C2000 Family (Piccolo and Delfino series)
- **Programming Languages**: Python, C, C++, Embedded C

Professional Experience

Intern–Tesla Energy Products Modeling and Integration **Dec 2025-May 2026**
Tesla Inc., Palo Alto, California

- Delivered grid code compliance and interconnection studies for two utility-scale BESS projects (300-500 MW); tuned PSCAD EMT models for project-specific requirements and benchmarked EMT simulations against CHIL results
- Developed load smoothing control functionality for GFM energy storage systems, integrating control logic into PSCAD EMT models to achieve up to 95% reduction in grid-seen load variability under utility defined ramp rate constraints
- Supported system level testing and model validation using controller hardware in the loop (CHIL) setups, benchmarking and improving simulation model fidelity to accurately reflect Megapack (Tesla BESS) product behavior

PhD Intern for Grid-Connected Converter Control **May-Aug 2024**
National Laboratory of the Rockies (formerly NREL), Golden, Colorado

- Performed EMT studies of GFM inverter systems, showing where classical Equal Area Criterion (EAC) and Lyapunov energy function-based methods mis-predict stability under current limiting and damping; quantified impact via CCT/CCA
- Built multi-inverter models in PLECS and assessed how non-uniform damping affects stability margins under faults
- Developed a circuit-equivalent energy-function for IBRs, yielding 30% less conservative stability predictions than EAC

Technical Power Consulting Intern **May-Aug 2022**
Hitachi Energy, Raleigh, NC

- Conducted interconnection studies (power flow, short-circuit, N-1 contingency) in PSSE for transmission-level systems (115-345 kV); automated case builds and post-processing via Python scripts, redcucing study turnaround time by 40%
- Synthesized high-resolution (1-second) PV irradiance data to improve the microgrid frequency stability study
- Performed a transmission routing study in QGIS and introduced a metric to rank the possible transmission routes

Academic Research and Hardware Experience

Project 1: Interoperability and Current-limiting Control of GFM Inverters

- Integrated direct, indirect, and hybrid current limiters into GFM inverters with closed-loop control architecture
- Experimentally validated a parallel GFM inverter platform (3×5 kW, 208 V) featuring heterogeneous primary controls (droop, VSM, dVOC) and different current-limiters. Demonstrated robust interoperability and power sharing

Project 2: Fault Ride-through (FRT) for Series-connected GFM Inverters

- Developed novel homogeneous (virtual impedance) and heterogeneous (mixed GFM-GFL) FRT strategies
- Proposed singular-perturbation-based controllers, ensuring small-signal stability and enhanced transient response
- Conducted hardware experiments with three 1 kVA/ 90 V series-connected converters to compare current limiting performance, demonstrating faster fault recovery (>25% reduction in resynchronization time) compared to baseline controllers

Project 3: Voltage Balancing of GFM Inverters in Unbalanced Microgrid

- Estimated required positive, negative, and zero sequence currents to support balanced voltage in unbalanced grid
- Demonstrated that a 3-wire GFM inverter with decoupled positive- and negative-sequence controllers can result in less than 1% load voltage imbalance, replicating the voltage balancing capabilities of an ideal voltage source

Project 4: Control and Modeling of Synchro-converter–based Hybrid AC–DC Microgrids

- Developed grid-forming synchro-converter control by embedding synchronous-generator dynamics into power-electronic controllers to provide inertial and frequency response in hybrid AC–DC systems
- Designed a bidirectional interlinking converter with a modified droop-based controller to enable robust power sharing, frequency regulation, and loss minimization between AC and DC subgrids

Project 5: Interaction Energy–based Transient Stability Analysis of Multi-GFM Inverter Systems

- Developed an analytical transient stability framework for multi–GFM inverter systems, enabling characterization of fault induced angle dynamics in current limited operation and validated in MATLAB/Simulink
- Formulated an inverter-to-inverter interaction energy metric capturing nonlinear coupling in multi-GFM systems, demonstrating how asymmetric inertia, damping, and power-reference tuning reduce stability margins

Selected Publications

IEEE ECCE – Interaction Energy-Based Transient Stability Framework for Multi-GFM Inverter Systems **2025**
IEEE COMPEL – Demystifying GFM Large-signal Stability Assessment with Equivalent-circuit Models **2025**
IEEE PESGM – Is Equal Area Criterion Applicable for Transient Stability Assessment of GFM Inverters? **2025**
IEEE PESGM – Stability Analysis of Multi-inverter Systems with Heterogeneous Grid-Forming Dynamics **2025**
IEEE TPEL – Overcurrent Limiting in Grid-Forming Inverters: A Comprehensive Review and Discussion **2024**
IEEE ECCE – Fault Ride-through for Cascaded Inverters using Mixed GFM and GFL Control **2024**
IEEE PESGM – Voltage Balancing of Grid-forming Inverters in Unbalanced, Islanded Microgrids **2023**

Honors and Awards

Best Paper Award, IEEE Transactions on Power Electronics (First Place) **Oct 2025**
Best Paper Award, IEEE Power and Energy Society General Meeting, Austin, TX **July 2025**
Best Poster Award, UNIFI Consortium Annual Meeting, Tempe, Arizona **Jan 2025**
George J. Heuer, Jr. Ph.D. Endowed Graduate Fellowship, UT Austin **July 2024**
Professional Development Award, UT Austin **May 2023**
Wilson-Tayabali Family Fellowship and Jordan Baruch Fellowship, UT Austin **Jan 2023**
Charles M. Simmons Endowed Presidential Fellowship in Engineering, UT Austin **Jan 2021**
Institute Day Academic Excellence Award, NIT Tiruchirappalli, India **Feb 2019**
MITACS Globalink Research Fellowship, Govt. of Canada **Jan 2019**

Services

Working Group Member, IEEE IBR Interconnection (IBRI), UNIFI Specifications for GFM Inverters **2025-2026**
Reviewed 100+ papers in IEEE Transactions on Power Electronics, Power Systems, and IEEE conferences **2023-2026**
Developed open-source educational materials for UNIFI consortium on GFM inverters **2024-2025**
Graduate Teaching Assistant for Power Quality and Harmonics, Power System Apparatus and Lab **2021-2023**

Leadership Roles

OpenMinds 2024 NextGen Energy and Climate Leader, OpenMinds Inc. **2024-2025**
TEX-E Energy Innovation Fellow, Martin Trust Center for MIT Entrepreneurship **2023-2024**
President, Longhorn Energy Club (LEC), UT Austin **2023-2024**